PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The effectiveness of pretend medical play in improving children's health outcomes and wellbeing: A systematic review
AUTHORS	Abdul Rashid, Aneesa; Cheong, Ai; Ranita, Hisham; Shamsuddin, Nurainul; Roslan, Dalila

VERSION 1 – REVIEW

REVIEWER	Dr. Asher Moser Medical School for International Health, Faculty of Health Sciences, ben Gurion University of the Negev Israel
REVIEW RETURNED	05-Aug-2020

GENERAL COMMENTS	This paper by Rashid et al attempts to address a very interesting topic that all pediatricians have a gut feeling that this is the "right thing". However, the task to prove it utilizing modern statistical tools and methods has turned out to be much more difficult, as the authors have experienced. The main reason for this difficulty, as correctly pointed out, is the design and the biases involved when dealing with the reactions and behaviors of (mostly) preschool children. To this effect, I do not see much importance in the inclusion of 16 year-olds as subjects of this review, as the teen-aged individuals have a completely different array of responses to these encounters that cannot be grouped with the younger children. The fact that there are many strong biases affects the strength of the conclusions reached in the paper as the statistical analysis is rendered to be weak. Re the issue of Ethics: Although is clear from the design of the individual studies included in the analysis that this is an difficult issue (mainly due to the fact that there is rarely an atmosphere to have informed consents), I would expect the authors of this paper to add a paragraph relating to this matter and explaining the difficulties. As for the outcomes, the nature and description of the limitations of the study, as stated by the authors, renders the conclusion reached as "probable" at best. Given all that, this paper tries to summarize the impressions that most individual articles included, which is that overall this a positive experience. I suggest the necessary changes to make this publish worthy.

REVIEWER	Peter Weinstock Boston Children's Hospital, Boston MA
REVIEW RETURNED	27-Aug-2020

GENERAL COMMENTS	General

	Authors do a nice job providing initial evaluation of the literature surrounding medical play via "Teddy Bear Hospital" Concept. The paper surrounds a small sample size due to signifiant exclusion criteria limiting interpretation of results. Line 47 – Typo “medical” Background Page 5, Line 55 – Analysis is significantly limited to studies on healthy children only. Rationale for this unclear. As well, study is limited to only articles appearing in English which causes >95% exemption from initial pool. Methods Minor point: Not sure reproduction of the total PRISMA 2009 Checklist is necessary. May be sufficient to reference. Mentioned only briefly in the Discussion section, foreign language exclusion (n=1091) is significant (95.7%) with likely large impact on ultimate findings and conclusions. Any way to increase language inclusion would be of strong interest towards meaningful conclusions. Removal of studies with sick patients. Rationale unclear. 80% of remaining excluded. Remaining 11 from 1154. Small sample size of available data. Page 7, Line 3 – Mesh Heading. May want to include “Simulation” as more studies are being done in this arena for patient and families and may improve your search results. Results Page 9, Line 52: 11 of 49 studies total reviewed of 1154 results in small sample size excluding all non-English articles. Children’s Feelings: Mixed results. Children’s Behavior: Mixed results. Children’s Healthcare Knowledge: Improved, though few studies to prove. Page 11, Line 32: Would be of interest to better describe the interventions of each studies with regard to time, staffing, curricula. Any studies include responses by parents? Would be of interest to note. Discussion: Would benefit from further elaboration on conceptual underpinnings of TBH including potential impacts on psychological safety, anxiety mitigation, learning, preparedness etc. With comparative references discussed. Page 12, Line 20. Not sure what is meant by “global project”. Page 12, Line 54. Would be of interest to understand the etiology of the detrimental effect described by Burns-Nader Page 14, Line 11. May want to include lines like this within the dedicated “Limitations” section. Authors may want to touch on the role of simulation pedagogy as

	there is growing literature around simulation-based interventions to prepare patient/families/caregivers among multiple clinical domains. Other May be worth including/commenting on: https://pubmed.ncbi.nlm.nih.gov/31243158/

REVIEWER	Mellanye Lackey University of Nevada Las Vegas Health Sciences Library
REVIEW RETURNED	22-Sep-2020

GENERAL COMMENTS	This reviewer suggests: -describing the language limitation as 'non-English' rather than foreign language. -PRISMA is a reporting checklist only, not for conducting the review as described in the manuscript. -The PRISMA checklist is not needed as a supplementary file since it is a publicly available document not specific to this review. - Instead include the specific search strategies for each database, the number of results from each search, along with specifics about the database. - The reported search strategy does not result in the total number of studies (1154). This reviewer copied and pasted the search strategy into PubMed.gov, added the language, date and publication limits, with 6430 results. This reviewer suggests checking the parentheses and boolean logic for accuracy. Currently, the nesting parentheses do not match. - This reviewer does not suggest using the limits feature in databases to exclude by language or publication type, since doing so eliminates items that have not been indexed yet. It is better to manually exclude them during the title/abstract screening instead. - The search strategy is flawed with the inclusion of an asterisk and double quotes. If an asterisk is added to the end of a phrase, truncation is performed. No double quotes are required around the phrase, as the asterisk acts as a phrase marker as well as truncation. For more information, see this article: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5624431/ - Additionally for the total number of results reported, this reviewer is skeptical about the number of duplicates reported (14), as PubMed, CINAHL, and Scopus each contain the MEDLINE database. If they were searched separately, then citations would be present in triplicate. Please clarify the search strategy and citation count. - This reviewer is also skeptical that Google Scholar was searched from inception, since a search for only teddy bear hospital in google scholar returned >33,000 citations, many more than the total number reported. - This reviewer questions why systematic reviews were excluded, since they would highlight relevant citations for and confirm findings of the current review.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1		
	This paper by Rashid et al attempts to address a very interesting topic that all pediatricians have a gut feeling that this is the "right thing". However, the task to prove it utilizing modern statistical tools and methods has turned out to be much more difficult, as the authors have experienced. The main reason for this difficulty, as correctly pointed out, is the design and the biases involved when dealing with the reactions and behaviors of (mostly) preschool children.	Thank you for your review. We agree we faced challenges in this manuscript write up as per your comments
	To this effect, I do not see much importance in the inclusion of 16 year-olds as subjects of this review, as the teen-aged individuals have a completely different array of responses to these encounters that cannot be grouped with the younger children.	We agree with your point. There was only one study by Morel et al that we had added during the update of our SR in 2020. This involved a wide range of children age ranging from 4-16 years old. Hence, we have omitted the study from this review as it can cause potential bias. Therefore, there are only 10 studies involved in this review Page 10, first paragraph All other sections highlighted in terms of reporting final results changing the review from 11 to 10 articles
	The fact that there are many strong biases affects the strength of the conclusions reached in the paper as the statistical analysis is rendered to be weak.	We hope the previous points have addressed the weakness
	Re the issue of Ethics: Although is clear from the design of the individual studies included in the analysis that this is an difficult issue (mainly due to the fact that there is rarely an atmosphere to have informed consents), I would expect the authors of this paper to add a paragraph relating to this matter and explaining the difficulties.	We have added this information in 'Limitations'. Page 13- second paragraph
	As for the outcomes, the nature and description of the limitations of the study, as stated by the authors, renders the conclusion reached as "probable" at best.	We do agree there are several limitations to this review. However, we hope that looking into this concept can help researchers and educators alike when doing educational health interventions on well children.
	Given all that, this paper tries to summarize the impressions that most individual articles included, which is that overall this a positive experience.	Thank you

	I suggest the necessary changes to make this publish worthy.	We have done our best to improve the quality of this manuscript based on your constructive recommendations. Thank you
	Reviewer 2	
	Authors do a nice job providing initial evaluation of the literature surrounding medical play via "Teddy Bear Hospital" Concept. The paper surrounds a small sample size due to signifiant exclusion criteria limiting interpretation of results.	Thank you for your kind comments- we have done our best to look into this concept at it has been used a popular tool worldwide.
	Line 47 – Typo “medical”	Thank you for pointing out this typo, we have made corrections
	Background Page 5, Line 55 – Analysis is significantly limited to studies on healthy children only. Rationale for this unclear. As well, study is limited to only articles appearing in English which causes >95% exemption from initial pool.	Since, most of the TBH done are done to children that are healthy such as kindergartens, or an open day in a specific health care Centre as an interventional health education; hence decided to exclude sick children. Page 5, last paragraph We have also added this information in the ‘exclusion’ section: As mentioned earlier, since most of the TBH done are done to children that are healthy such as kindergartens, or an open day in a specific health care center as an interventional health education; hence we excluded sick children. Furthermore, the psychology of sick hospitalised children will most likely differ to those that are not.[13] Page 7, 1st paragraph Thank you for your suggestion. We have revised our search strategy according to reviewers’ comments and we have included non-English articles in the search (search strategy: supp file S1, PRISMA Flow: chart fig 1)
	Methods Minor point: Not sure reproduction of the total PRISMA 2009 Checklist is necessary. May be sufficient to reference.	We agree with your suggestion and have omitted this file. Page 6, methods.
	Mentioned only briefly in the Discussion section, foreign language exclusion (n=1091) is significant (95.7%) with likely large impact on ultimate findings and conclusions. Any way to increase language inclusion would be of strong interest towards meaningful conclusions. Removal of studies with sick patients. Rationale unclear. 80% of remaining excluded. Remaining 11 from 1154. Small sample size of available data. Page 7, Line 3 – Mesh Heading. May want to include “Simulation” as more	We have revised our search strategy according to reviewers’ comments and we have included non-English articles in the search. (search strategy: supp file S1, PRISMA Flow: chart fig 1) We have added in the rationale as earlier requested Thank you for your suggestion. We have revised our search strategy based on the reviewer’s comments. However, there was no Mesh heading

	studies are being done in this arena for patient and families and may improve your search results	for 'Simulation' that were relevant to our study objectives. After discussing with the team, we found that Mesh heading "Play and Playthings" were more relevant and we have included it in our search strategy. (search strategy: supp file S1, PRISMA Flow: chart fig 1)
	Results Page 9, Line 52: 11 of 49 studies total reviewed of 1154 results in small sample size excluding all non-English articles. Children's Feelings: Mixed results. Children's Behavior: Mixed results. Children's Healthcare Knowledge: Improved, though few studies to prove.	 We have made big changes due to revision of the search strategy on this section Page 9-Results Thank you for your input we have added in the results section to make it clearer for the readers.
	Page 11, Line 32: Would be of interest to better describe the interventions of each studies with regard to time, staffing, curricula. Any studies include responses by parents? Would be of interest to note.	We have added the interventions regarding time, staffing and curricula Page 10, first paragraph We have also added this information in the supplementary file 4 (S4) We have added in this note on parents in "results of individual studies"; "others" Page 11-last paragraph
	Discussion: Would benefit from further elaboration on conceptual underpinnings of TBH including potential impacts on psychological safety, anxiety mitigation, learning, preparedness etc. With comparative references discussed. Page 12, Line 20. Not sure what is meant by "global project". Page 12, Line 54. Would be of interest to understand the etiology of the detrimental effect described by Burns-Nader Page 14, Line 11. May want to include lines like this within the dedicated "Limitations" section. Authors may want to touch on the role of simulation pedagogy as there is growing literature around simulation-based interventions to prepare patient/families/caregivers among multiple clinical domains	Thank you, we have added in our discussion pertaining to your suggestion. Page 12- last paragraph We have edited this part to "a popular project that have been done around the world" Page 12-2nd paragraph This involved reported increase fear and lower mood by the children after the play therapy and after the doctor's visit compared to baseline Page 13-last paragraph We have added this part. Page 15- 2nd paragraph Thank you for the suggestion. We have added this information Page 13 -first paragraph. However, the term "experiential learning" were more relevant to our study objective compared to "simulation" where it was mostly used in teaching medical education.

	Reviewer 3	
	This reviewer suggests: -describing the language limitation as 'non-English' rather than foreign language.	Thank you for your suggestion. We have revised our search strategy according to reviewers' comments and we have included non-English articles in the search. (search strategy: supp file S1, PRISMA Flow: chart fig 1)
	-PRISMA is a reporting checklist only, not for conducting the review as described in the manuscript. -The PRISMA checklist is not needed as a supplementary file since it is a publicly available document not specific to this review.	Thank you for your suggestion, we have omitted the check list
	- Instead include the specific search strategies for each database, the number of results from each search, along with specifics about the database.	Thank you for your comment. We have revised the search strategy. The search strategy and number of results is included in the Supplementary file 1. (search strategy: supp file S1, PRISMA Flow: chart fig 1)
	The reported search strategy does not result in the total number of studies (1154). This reviewer copied and pasted the search strategy into PubMed.gov, added the language, date and publication limits, with 6430 results. This reviewer suggests checking the parentheses and boolean logic for accuracy. Currently, the nesting parentheses do not match.	Thank you for your comment. We have found that there was an error in reporting the initial search number of results found. It is not 1154 but 3139. However, based on reviewers' comments we have revised our search strategy. We have also considered all the technical search issues such as additional keywords, MeSH headings, Boolean logic and parentheses. The total number found is 16,306.
	- This reviewer does not suggest using the limits feature in databases to exclude by language or publication type, since doing so eliminates items that have not been indexed yet. It is better to manually exclude them during the title/abstract screening instead.	Based on the reviewers' comments, we have included all non-English articles and publication types.
	- The search strategy is flawed with the inclusion of an asterisk and double quotes. If an asterisk is added to the end of a phrase, truncation is performed. No double quotes are required around the phrase, as the asterisk acts as a phrase marker as well as truncation. For more information, see this article: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5624431/	Thank you for the comment and guide. We agree and have revised the search strategy as suggested.
	- Additionally, for the total number of results reported, this reviewer is skeptical about the number of duplicates reported (14), as PubMed, CINAHL, and Scopus each contain the MEDLINE database. If they were searched separately, then citations	Thank you for your comment. We have found that there was an error in reporting the initial search number of results found. We have revised the search strategy and removed duplicates. The numbers are reported in the PRISMA flowchart (fig 1)

	would be present in triplicate. Please clarify the search strategy and citation count.	
	- This reviewer is also skeptical that Google Scholar was searched from inception, since a search for only teddy bear hospital in google scholar returned >33,000 citations, many more than the total number reported.	We understand that searching in Google Scholar can retrieve more than 33,000 articles. However, using our search strategy, we managed to reduce the search results. For Google Scholar, our search was based on previous published papers that takes the first 200 relevant articles to be included in the search results, due to large number of results found (as demonstrated above). Link to articles are here: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-017-0644-y https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0138237#sec001
	- This reviewer questions why systematic reviews were excluded, since they would highlight relevant citations for and confirm findings of the current review	Initially, when we mentioned on exclusion of systematic reviews, we intended to look for the studies mentioned in the review individually by retrieving the individual full text and citation tracking. However, due to this issue, we have decided to remove systematic reviews as the exclusion criteria.

VERSION 2 – REVIEW

REVIEWER	Peter Weinstock MD PhD Boston Children's Hospital, Boston MA USA
REVIEW RETURNED	14-Dec-2020

GENERAL COMMENTS	The authors as done a nice job responding to suggestions and comments.